PORTFOLIO PROJECT · HEALTHCARE ANALYTICS

Hospital Appointment No-Show Analysis

Why patients miss appointments, and what a hospital can do about it

110,527
appointment records

62,299
unique patients

25
columns after feature engineering

Alfahad K
AF05361543

PROBLEM STATEMENT

Empty slots: the hospital's hidden cost

THE PROBLEM

Patients book appointments and do not attend. Each no-show wastes a slot that another patient could have used.

THE GOAL

Find out why patients miss appointments, and recommend actions that raise attendance.

What no-shows damage

- Scheduling and resource use
- Doctor availability
- Waiting times for other patients
- Operational efficiency
- Quality of care

KEY QUESTIONS

Why do patients miss? Which ages, days and areas? Does waiting time or an SMS reminder matter? What should we do?

HOW I INVESTIGATED

110,527 appointments, one clear workflow

01 CLEAN

Python, Pandas

Fix dates, code no_show 0 or 1, add age group, waiting days and weekday

02 STORE

MySQL

25 columns, 110,527 rows imported, 0 skipped

03 ANALYSE

SQL

No-show rate by age, weekday, waiting time, SMS, area, health condition

04 REPORT

Power BI

Three-page dashboard with slicers

Data: 110,527 appointments, 62,299 unique patients, Brazil 2016 (public Medical Appointment No Shows dataset).

```
no-show rate = SUM(no_show) × 100 / COUNT(*)
```

0 = attended, 1 = missed. Counts appointments, not patients.

WHAT THE ANALYSIS FOUND

Six patterns behind the missed appointments

WAITING TIME

Longer waits, more no-shows

Patients who waited longer between booking and visit were more likely to miss it.

WEEKDAY

Some days lose more slots

No-show rates differ across weekdays.

AGE GROUP

Some ages miss more

Certain age segments were more likely to miss.

SMS REMINDERS

Attendance differs by reminder

Different trends. An association, not proof of cause.

NEIGHBOURHOOD

Location matters

Several neighbourhoods consistently had higher no-show rates.

HEALTH CONDITIONS

Chronic conditions differ

Hypertension and diabetes patients showed different attendance patterns.

THE SOLUTION

Each cause gets a targeted action

Cause found	Recommended solution	Track
Long waiting time	Shorten the gap between booking and visit	Wait days, no-show rate
Weekday differences	Adjust reminder schedule and staffing by day	No-show rate by weekday
Higher-risk age groups	Age-appropriate communication and engagement	No-show rate by age group
Mixed SMS results	Expand SMS reminders and review timing and coverage	Attendance, SMS vs no SMS
High no-show neighbourhoods	Investigate local barriers and add outreach	No-show rate by area
Chronic conditions	Targeted engagement for high-risk patients	Attendance by condition

TOGETHER Fewer empty slots, shorter queues and better use of doctors' time, monitored on a KPI dashboard.

IMPACT AND NEXT STEPS

How the hospital would know it is working

EXPECTED IMPACT

- Fewer wasted appointment slots
- Better doctor availability
- Shorter waits for other patients
- Smoother daily operations

NEXT STEPS

- Pilot reminders and measure the change
- Build a no-show risk model with a baseline
- Keep Power BI KPIs live for monitoring

LIMITS

- 2016 data may not match today
- Patterns, not proven causes
- Descriptive, no prediction model

HOSPITAL APPOINTMENT NO-SHOW ANALYSIS

Thank you

Questions and feedback are welcome.

EMAIL
alfahad890688@gmail.com

LINKEDIN
linkedin.com/in/alfahadk

GITHUB
github.com/alfahadk

PORTFOLIO
alfahadk-portfolio.netlify.app

Alfahad K · Data Analyst